Long-Term Trends and Mesoscale Eddies Impact on Hyperspectral Particulate Absorption Coefficients in the South China Sea

Wenlong Xu, Yufan Yin, Guifen Wang, Xiaogang Xing, Biao Jin, Alex Hayward, Marin Cornec, and Emanuele Organelli

***Abstract*—Documenting the variations of hyperspectral particulate absorption coefficients is essential for understanding how light is propagated underwater and thereby estimating marine primary production. Using an extensive in situ dataset (2004–2018), this study investigates the spatial, vertical, and long-term trends of hyperspectral particulate absorption coefficients in the South China Sea (SCS). The dataset comprises 608 profiles (spanning depths from surface to 100 m) and includes measurements of the spectral absorption coefficients for total particulates, phytoplankton, and nonalgal fractions between 2004 and 2018. Our results indicate that particulate absorption coefficients at 443 nm (chlorophyll-a peak) exhibit a clear decreasing gradient from the coastal to open ocean, varying by two orders of magnitude (from ∼0.005 to ∼0.5 m^{-1}), with maxima typically observed in the subsurface layer in the open ocean. The influence of mesoscale eddies on particulate absorption was subtle at the surface but became more pronounced with depth, where both the magnitude and spectral shape differed increasingly between anticyclonic and cyclonic eddies. The observational results reveal an interdecadal shift in particulate absorption coefficients around 2009, characterized by a nonsignificant increasing trend from 2004 to 2009 (Slope = 0.0042 m^{-1} yr^{-1}, P = 0.12) followed by a significant decline from 2009 to 2018 (Slope = −0.0016 m^{-1} yr^{-1}, P = 0.05) at 75 m depth. This variability shows a marginally significant negative correlation with the Pacific Decadal Oscillation (PDO) (R = −0.52, P = 0.07). This assessment of hyperspectral particulate absorption coefficients establishes a foundation to refine ocean-color products in the SCS.**

***Index Terms*—Long-term trends, mesoscale eddies, particulate absorption coefficient, the South China Sea (SCS).**

NOMENCLATURE

a_p	Total particulate absorption coefficient (m^{-1}).
a_{ph}	Phytoplankton absorption coefficient (m^{-1}).
a_{nap}	Nonalgal particulate absorption coefficient (m^{-1}).
Snap	Spectral slope of a_{nap}(nm^{-1}).
λ	Wavelength (nm).
MLD	Mixed layer depth (m).
SST	Sea surface temperature (°C).
$Z_{\sigma=23}$	Depth of potential density at 23 kg m^{-3}(m).
$Z_{T=22}$	Depth of temperature at 22 °C (m).

I. INTRODUCTION

LIGHT absorption is a key optical property of natural waters that regulates the propagation of the underwater light field [1], [2], [3], [4], [5], [6] and serves as a spectral filter for incident and scattered solar radiation [7], [8]. Light absorption coefficients of optically active components (e.g., phytoplankton, sediments) are among the most important parameters for characterizing the variability of natural waters and their apparent optical properties [9]. As such, accurate characterization of light absorption coefficients and their variability is essential for understanding numerous physical and biological processes in the upper ocean. These bio-physical processes are complex and are influenced by the magnitude of solar energy, including photochemical reactions, water-column heating, photosynthesis, and light availability [5], [7]. Variations in the spectral shapes of the absorption coefficients can also indicate the composition and concentration of nonwater constituents such as phytoplankton pigments, community structure, and particle size [5], [6], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Furthermore, accurate absorption coefficient data are fundamental for estimating marine primary production [19], [20], [21]. Therefore, the spectral absorption coefficients of particulate matter (a_p) and its two main components—phytoplankton (a_{ph}) and nonalgal particles (a_{nap})—have been extensively investigated in the

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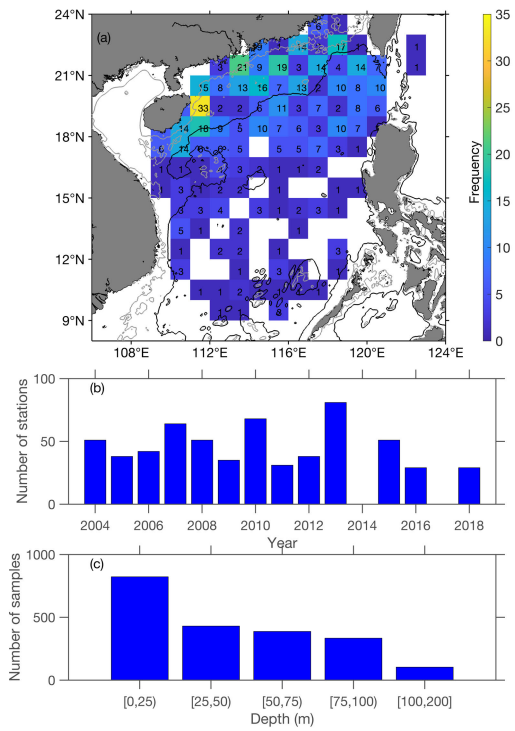


Fig. 1. Spatial and temporal distribution of sampling stations (total of 645 stations) from 2004 to 2018, acquired during 15 cruises. (a) Frequency distribution of sampling stations in each $1^\circ \times 1^\circ$ grid cell during the 2004–2018 period. Bathymetry contours show 50 (gray) and 1000 (black) m isobaths, which were obtained from ETOPO2. (b) Annual number of sampling stations. (c) Frequency distribution of sampling stations by sampling depth.

global ocean over the past three decades [3], [9], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]. However, many oceanic and coastal regions remain poorly characterized, and the magnitude, variability, and long-term trends of absorption properties in relation to oceanic physical processes are still not well understood, one of these areas being the South China Sea (SCS).

The SCS is the largest semi-enclosed sea in the western tropical Pacific ($\sim 3.5 \times 10^6$ km^2) and features a deep Northeast–Southwest basin connected to surrounding seas via several straits, including the Luzon, Mindoro, Taiwan, and Karimata Straits [Fig. 1(a)]. The Pearl River Estuary lies along the northern boundary of the SCS, adjacent to the broad continental shelf. The SCS is strongly influenced by the East Asian monsoon system, with prevailing southwesterly winds in summer (May–August) and northeasterly winds in winter (October–April) [32], [33], [34]. Meanwhile, the Kuroshio intrusion exerts a significant influence on the upper-layer circulation of the SCS, particularly in the northern region and along the western boundary [35], [36], [37].

The SCS exhibits highly complex topography and dynamic oceanographic conditions, characterized by active mesoscale eddies [33], [38], [39], frequent upwelling events [40], [41], and pronounced frontal systems [42], [43], [44]. From the ecosystem and bio-optical perspective, the open SCS shares many features with other oligotrophic oceanic regions, such as low nitrate availability in the surface waters [45]. Consequently, numerous studies have investigated the regional, seasonal, and long-term variability of physical processes and their impacts on biogeochemical dynamics [44], [46], [47], [48], [49]. However, research studies focusing on the bio-optical properties of the SCS remain limited. Previous efforts, based on sparse in situ observations, have primarily aimed at evaluating remote sensing algorithms [50], decomposing bulk absorption coefficients into their constituent components [25], deriving phytoplankton pigment concentrations from absorption spectra [51], or assessing the influence of physical processes on bio-optical variability [52], [53], [54], [55]. In contrast, the spatial, vertical, and long-term variability of particulate absorption coefficients, as well as their responses to mesoscale eddies, remains largely unexplored.

The aims of this article are as follows.

1) To describe the spatial and vertical variations and long-term trends of hyperspectral particulate absorption coefficients in the SCS.

2) To investigate the impact of mesoscale eddies on these absorption properties. The remainder of this article is organized as follows. Section II describes the data and methods. The results and discussion are presented in Section III, covering the observed spatial variability, the effects of mesoscale eddies, relationships with environmental factors, and long-term trends. A summary of the findings is provided in Section IV. Nomenclature section summarizes the symbols and abbreviations used in this study.

II. Data and Methods

A. Study Area and Sampling

Samples were collected during 15 cruises between 2004 and 2018 in the SCS (Fig. 1 and see Supplemental Table SI). Samples were collected aboard R/V "Shiyan 1," "Shiyan 3," and "TAN KAH KEE." The sampling stations spanned the entire SCS, from coastal to open ocean regions, with the most samples collected in the northern SCS [Fig. 1(a)]. The annual distribution of sampling stations is shown in Fig. 1(b) and varied among years. A minimum of 25 stations were sampled each year, except in 2014 and 2017, when no field observations were conducted due to logistical constraints. At each station, water samples were collected at standard depths of 5, 25, 50, 75, and 100 m. At selected open ocean stations, additional samples were taken at 150 and 200 m to capture deeper optical variability. Owing to shallow local water depths and operational limitations at many stations, most samples were collected within the upper 25 m of the water column [Fig. 1(c)].

B. Particulate Absorption Measurements

For particulate absorption measurements, seawater samples (with volumes varying from 1 L in the coastal area to 4 L in the open ocean) were filtered onto Whatman GF/F filters (25 mm in diameter). Prior to analysis, filters were stored in the dark at low temperature. Absorption spectra were measured directly on the filters using a Helios-γ, UV-2550 (Shimadzu), or Perkin-Elmer Lambda-650 spectrophotometer, with a blank wet filter as [56]. Spectra were recoded between 350 and 750 nm with 1-nm intervals. Phytoplankton pigments were extracted from filters using methanol for 90–180 min following [57]. Following pigment extraction, the sample was then reanalyzed to determine the nonalgal absorption spectra using the same procedure. All spectra were baseline-corrected by subtracting the optical density at 750 nm to minimize potential differences between sample and reference filters [9], [25]. Optical densities were further corrected for the path-length

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amplification effect, using the algorithms developed by [58]. The phytoplankton absorption spectra, $a_{ph}(\lambda)$, were obtained as the difference between total particulate absorption, $a_p(\lambda)$, and nonalgal particle absorption, $a_{nap}(\lambda)$. The consistency of absorption measurements obtained from different instruments used across multiple cruises was evaluated by comparing $a_{ph}(443)$ with concurrent chlorophyll-a measurements. The strong agreement shown in Fig. S1 demonstrates good interinstrument consistency and overall measurement reliability.

The $a_{nap}(\lambda)$ generally decreases exponentially with increasing wavelength [9], [24], [59] and can be described by the following equation:

$$a_{nap}(\lambda) = a_{nap}(\lambda_0) \exp\left(S_{nap}(\lambda_0 - \lambda)\right) \quad (1)$$

where λ_0 is the reference wavelength, and S_{nap} is the spectral slope of $a_{nap}(\lambda)$. Based on our dataset collected across diverse water types, $a_{nap}(\lambda)$ was modeled using this exponential relationship over the 400–700 nm spectral range, with 443 nm chosen as the reference wavelength.

C. Mixed Layer Depth

Vertical profiles of temperature and salinity were acquired using a ship-deployed conductivity–temperature–depth (CTD) system, with measurements spanning from the surface to 1000 m depth at a vertical resolution of 1 m. The variables used in this study, along with their corresponding symbols, definitions, and units, are summarized in the Nomenclature section. Mixed layer depth (MLD) was calculated as the depth where potential density exceeds the value at 10 m by 0.03 kg m^{-3}, following [60]. Comparisons with alternative definitions proposed by Suga et al. [61] and Holte and Talley [62] showed strong consistency, with correlation coefficients (R) of 0.84 and 0.82, respectively (Fig. S2). Given this agreement and its widespread use in upper-ocean studies, Weller and Plueddemann [60] adopted for subsequent analyses.

D. Mesoscale Eddy Trajectory Atlas

To identify eddy centers, boundaries, amplitudes, and radii in the study region, we used data from the Mesoscale Eddy Trajectory Atlas (META) [63]. The delayed-time all-satellite product, META3.2 (SSALTO/DUACS and AVISO+, 2022), was used to cover the period from 2004 to 2018. A key advancement in these datasets is the use of the absolute dynamic topography (ADT) for eddy detection [63]. Eddy identification was performed using the *py-eddy-tracker* algorithm developed by [64] and described in detail by [63]. The datasets and accompanying documentation are publicly available on the AVISO+ website (https://www.aviso.altimetry.fr).

To differentiate the impact of eddies on particulate absorption, in situ stations were categorized according to their distance (d) from the eddy center. Stations located within the eddy radius ($d \leq R$) were designated as eddy-influenced, whereas those beyond this threshold ($d > R$) were considered ambient (noneddy) waters.

E. Ancillary Data

To describe the spatial distribution of surface particulate absorption coefficient and long-term linear trends at the basin scale, the European Space Agency (ESA) ocean color climate change initiative (OC-CCI) phytoplankton absorption coefficient (a_{ph}) product was used [65]. The datasets are publicly available at https://www.oceancolour.org/. The ESA OC-CCI satellite products for a_{ph} were first validated using in situ bio-optical measurements collected in the SCS. Scatterplots comparing satellite-derived and in situ a_{ph} were shown in Fig. S3. The agreement was highest at shorter wavelengths (< 510 nm), with coefficients of determination (R^2) ranging from 0.616 to 0.725, root mean square deviation (RMSD) ranging from 0.004 to 0.012, and biases consistently below 0.0063 [Fig. S3(a)–(d)]. However, the correlation weakened considerably with increasing wavelength, becoming negligible at 665 nm [R^2 = 0.026, RMSD = 0.867, and BIAS = −0.6074; Fig. S3(f)]. Furthermore, a certain degree of underestimation by the satellite product was evident at 412 and 443 nm, particularly at higher a_{ph} values.

The Pacific Decadal Oscillation (PDO) index was obtained from the National Centers for Environmental Information (https://www.ncei.noaa.gov/pub/data/cmb/ersst/v5/index/ersst.v5.pdo.dat). This index is defined as the leading pattern empirical orthogonal function of sea surface temperature (SST) anomalies in the North Pacific basin, typically, polewards of 20°N. Further details can be found in [66] and [67]. The multivariate El Niño–Southern Oscillation index version 2 (MEI.v2) from the National Oceanic and Atmospheric Administration (NOAA) was used to characterize ENSO variability in this study. Further details on the index and its calculation are available on the NOAA website: https://www.psl.noaa.gov/enso/mei/

F. Statistical Analyses

Statistical analyses were conducted to examine the linear relationships between particulate absorption coefficients and environmental variables. Correlation strength and significance were evaluated using linear regression and a two-tailed Student's *t*-test implemented via the MATLAB test function.

III. Results and Discussion

A. General Variability of Particulate Absorption

The analysis focused on particulate absorption coefficients at 443 nm, a wavelength with well-established sensitivity to phytoplankton biomass, pigments, and colored detrital material [9], [12]. The $a_p(443)$ from the 15 cruises (Fig. 2) exhibited large variability, ranging from approximately 0.005 to 0.5 m^{-1}—spanning two orders of magnitude [Fig. 2(a)–(e)]. The phytoplankton absorption coefficient, $a_{ph}(443)$, also had a broad range (0.001–0.29 m^{-1}) [Fig. 2(f)–(j)]. The nonalgal particle absorption coefficients, $a_{nap}(443)$, averaged 0.0084 ± 0.0174 m^{-1} and ranged from 0.0001 to 0.34 m^{-1}. Both $a_p(443)$, $a_{ph}(443)$, and $a_{nap}(443)$ were higher closer to the coastal (0.1887 ± 0.3306 m^{-1}, 0.0886 ± 0.1401 m^{-1}, and 0.1007 ± 0.2591 m^{-1}, respectively) and lower in the open ocean (0.029 ± 0.0375 m^{-1}, 0.0213 ± 0.17 m^{-1}, and 0.0077 ± 0.0276 $m-^{-1}$, respectively), with pronounced spatial variability across the study region.

To further examine the coastal/open ocean and vertical variability of hyperspectral particulate absorption coefficients, the in situ dataset was separated by sampling depth and coastal/open ocean configuration (defined following the station bathymetry) (Fig. 3). In coastal waters with bottom depths < 50 m, the mean particulate absorption spectra (a_p and a_{ph}) decreased with depth, with the highest values consistently observed near the surface [Figs. 2 and 3(a) and (b)], likely

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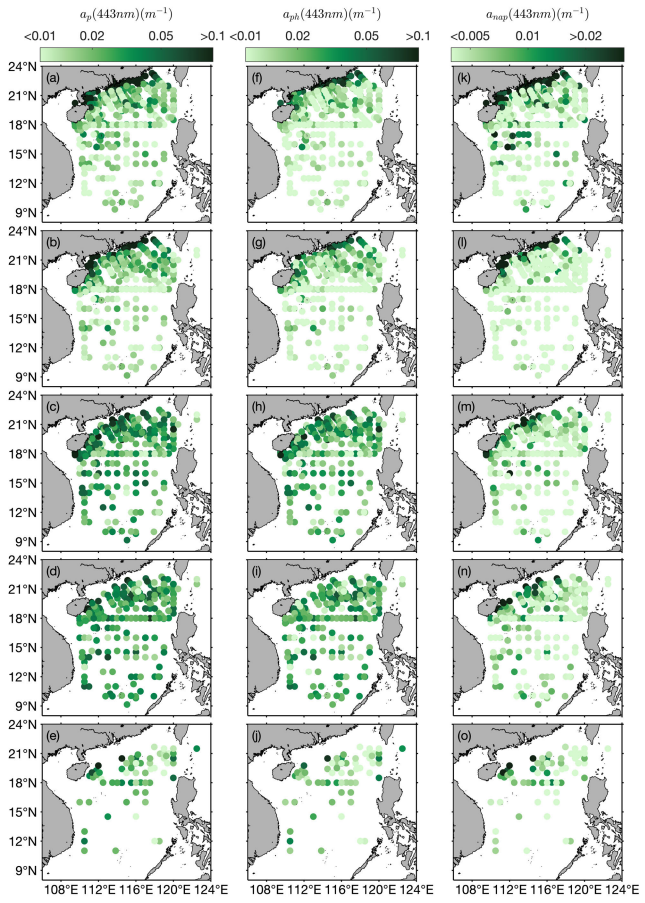


Fig. 2. Spatial and vertical variations in particulate absorption coefficient at 443 nm in the SCS. (Left) Total particulate absorption coefficient (a_p). (Middle) Phytoplankton absorption coefficient (a_{ph}). (Right) Nonalgal particulates absorption coefficient (a_{nap}). (a), (f), and (k) 5 m. (b), (g), and (l) 25 m. (c), (h), and (m) 50 m. (d), (i), and (n) 75 m. (e), (j), and (o) 100 m.

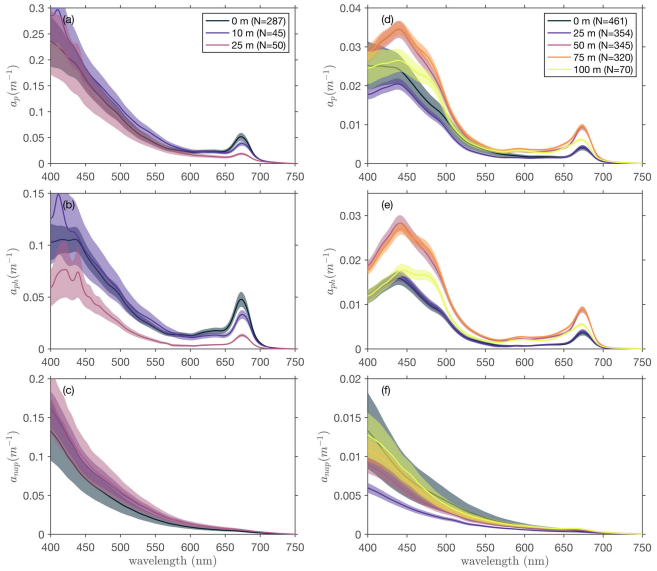


Fig. 3. Average vertical variations in hyperspectral particulate absorption coefficients (a) and (d) a_p, (b) and (e) a_{ph}, and (c) and (f) a_{nap} in the SCS coastal and open ocean waters. (a)–(c) (Left) Coastal ocean stations with water depth less than 50 m. (d)–(f) (Right) Open ocean stations with water depth up to 100 m. Different color lines represent the average values for different sampling depths (5, 10, 25, 50, 75, and 100 m), and the shaded colors show the standard deviations. N indicates the number of samplings.

reflecting elevated phytoplankton biomass and terrigenous particulate inputs. And the average of a_{nap} increased with depth, with lowest values presented at the bottom [Figs. 2 and 3(c)]. In contrast, in open ocean waters (deeper than 50 m), the mean particulate absorption spectra (a_p, a_{nap}, and a_{ph}) generally increased with depth, with highest values typically occurring around the subsurface chlorophyll maximum (SCM, 50 or 75 m), and then decreased [Fig. 3(d)–(f)]. The spectral shape of particulate absorption differs markedly between coastal and open ocean waters. A strong exponential decay in coastal spectra reflects the dominance of nonalgal particles, which constitute ~53% of a_p at 443 nm. In the open ocean, the spectral shape is marked by a phytoplankton signature, where a_{ph} accounts for a larger proportion (~73%) of absorption at 443 nm.

The spectral slopes of nonalgal particle absorption (S_{nap}) were calculated for the coastal and open ocean stations and different sampling depths, yielding a mean value of 0.012 nm^{-1} and ranging from 0.005 to 0.035 nm^{-1}. A subtle yet significant regional gradient was observed in S_{nap}, with coastal waters (0.0126 ± 0.002 nm^{-1}) exhibiting slightly higher values than open ocean waters (0.0122 ± 0.002 nm^{-1}). The elevated coastal S_{nap} is consistent with the influence of terrestrial inputs and a particle assemblage characterized by smaller, more absorbent particles [68].

It is important to note that particulate absorption coefficients exhibit not only strong spatial and vertical variability but also pronounced seasonal variations, driven by changes in phytoplankton community composition, biomass, and particulate matter inputs [69], [70], [71]. However, the present study focuses exclusively on data collected during the summer months (June–September, Table S1), as adverse sea conditions and logistical limitations prevented sampling during other seasons. As a result, potential seasonal patterns in particulate absorption remain unresolved in the SCS. Future multiseasonal field campaigns are needed to capture these dynamics and develop a more comprehensive understanding of temporal variability in the SCS.

B. Relationship Between Particulate Absorption Coefficients and Environmental Factors

To investigate the drivers of particulate absorption coefficients variability, we analyzed their relationship with key environmental factors in the open ocean (Fig. 4). Statistical analysis revealed that a_p, a_{ph}, and a_{nap} at 443 nm were only weakly correlated with factors across the water column such as SST, MLD, the depth of potential temperature at 22 °C ($Z_{t=22}$), representing the thermocline in the SCS, and the depth of potential density at 23 kg m^{-3} ($Z_{\sigma=23}$), representing the pycnocline in the SCS. These weak correlations suggest that the dynamics controlling particulate absorption are complex. Nevertheless, SST, $Z_{t=22}$, and $Z_{\sigma=23}$ emerged as the most significant potential drivers, showing weak correlations that are marginally statistically significant ($R < 0.35$, $P < 0.1$). Notably, we found that $a_p(443)$, $a_{ph}(443)$, and $a_{nap}(443)$ at 25 m depth had stronger correlations with SST. Specifically, a_p, a_{ph}, and a_{nap} at 443 nm showed a significant negative

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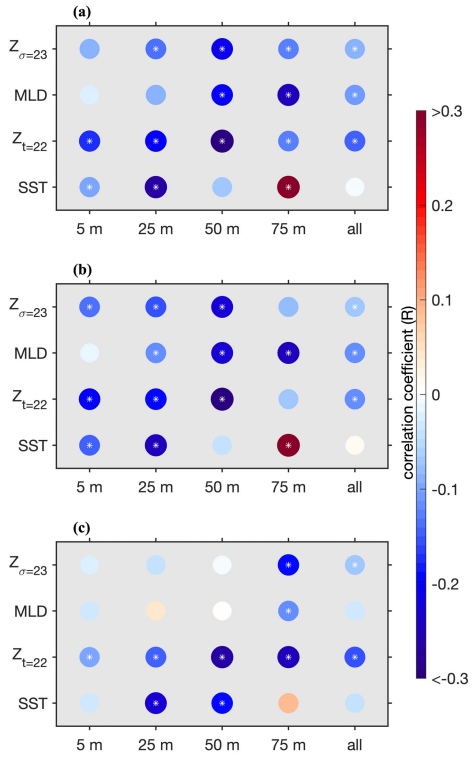


Fig. 4. Relationships between particulate absorption coefficients at 443 nm and environmental factors in the open ocean. Both marker size and shading color represent the values of correlation coefficient (−0.3 to 0.3). Stippling in (a)–(c) indicates where the correlations are significant at the 90% confidence level.

correlation with SST [R = −0.27, P < 0.001; R = −0.25, P < 0.001; R = −0.23, P < 0.001; respectively, Fig. 4(a)–(c)]. In contrast, at 75 m depth, a_p and a_{ph} exhibited a significant positive correlation with SST [R = 0.30, P < 0.001; R = 0.32, P < 0.001, respectively, Fig. 4(a) and (b)].

Overall, particulate absorption coefficients in the upper 50 m were negatively correlated with SST, suggesting that phytoplankton pigments or biomass decline as SST rises, likely due to enhanced stratification shallowing the MLD, thereby limiting nutrient availability [72]. This trend is expected to strengthen under future global warming scenarios. It is important to note, however, that this relationship is based solely on summer in situ observations and may not fully represent other seasons, which could weaken or alter the correlation. Therefore, establishing a more robust understanding will require incorporating observations from multiple seasons in future studies.

C. Eddies Effect on Particulate Absorption Coefficients

Mesoscale eddies are known to play a pivotal role in regulating phytoplankton dynamics by altering horizontal advection, vertical nutrient fluxes, and upper-ocean stratification [49], [73], [74], [75], [76], [77], [78], [79]. ACEs generally deepen the pycnocline, while CEs shoal it, thereby influencing nutrient availability and light conditions that shape biological productivity. Given that particulate absorption spectra are intrinsically linked to phytoplankton concentration and composition, we hypothesize that these eddy-driven physical–biogeochemical interactions significantly influence particulate absorption throughout the water column.

TABLE I
DESCRIPTIVE STATISTICS OF EDDY CHARACTERISTICS

	Mean	Median	25th percentile	75th percentile	Minimum	Maximum
Amplitude AE (cm)	8.2	8.1	4.2	12.1	0.7	18.6
Amplitude CE (cm)	6.6	5.1	2.8	10.8	0.8	17.6
Radius AE (km)	128.3	135.9	89.4	150.4	41.6	238.9
Radius CE (km)	97.7	91.7	67.5	123.1	34.5	178.8

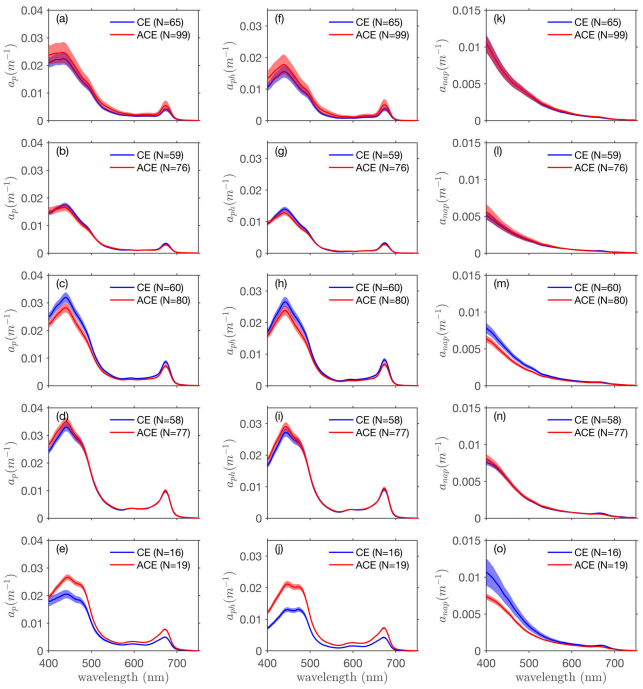


Fig. 5. Observed eddies influence on hyperspectral particulate absorption coefficients. (a)–(e) (Left) Total hyperspectral particulate absorption coefficients (a_p), grouped by the cyclonic eddies (blue lines) and anticyclonic eddies (red lines), respectively. (f)–(j) (Middle) and (k)–(o) (Right) Absorption spectra of phytoplankton (a_{ph}) and nonalgal particulates (a_{nap}), respectively. The color shadings indicate one standard deviation, and the lines represent the mean values. N indicates the number of samplings.

1) Eddies Characteristics: The impact of mesoscale eddies on particulate absorption was also analyzed. Eddy characteristics are detailed in Table I. Amplitudes ranged from ~0.7 to > 18.6 cm, with mean values of 6.6 cm for CEs and 8.2 cm for ACEs. Radii spanned 34.5–238.9 km, with median values of 97.7 and 128.3 km for CEs and ACEs, respectively..

2) Observed Vertical Patterns and Underlying Mechanisms: The particulate absorption coefficients observed within the eddies (Fig. 5) reveal subtle differences between CEs and ACEs. Although the shape of mean absorption spectra was similar, vertical disparities were evident (Fig. 5). Interestingly, CEs did not consistently enhance particulate absorption coefficients, nor did ACEs always suppress them. At 50 m depth, a_p, a_{ph}, and a_{nap} in the CEs were slightly higher than in the ACEs [Fig. 5(c), (h), and (m)]. In contrast, at 75 m depth, a_p

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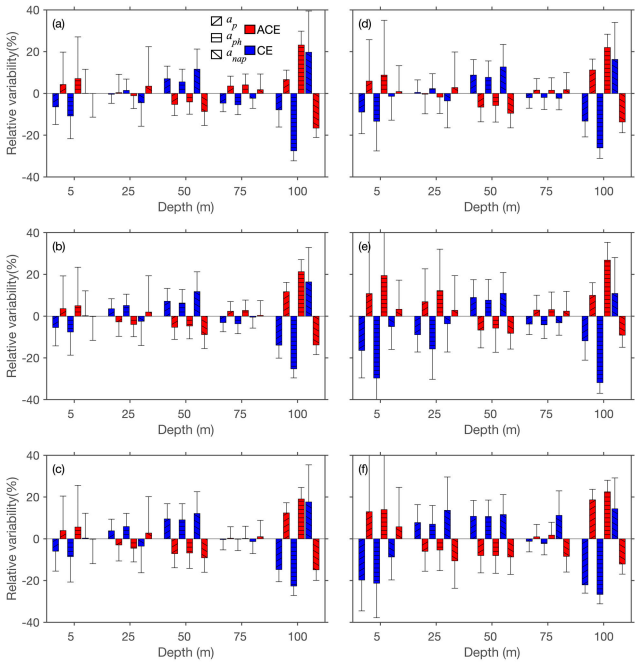


Fig. 6. Relative variability in mesoscale impact on particulate absorption coefficients in six bands [(a) 412, (b) 443, (c) 490, (d) 510, (e) 560, and (f) 665 nm] for anticyclonic (red bars) and cyclonic (blue bars) eddies in the SCS, with standard deviations labeled atop the respective bars. (a)–(f) Correspond to different wavelengths, respectively.

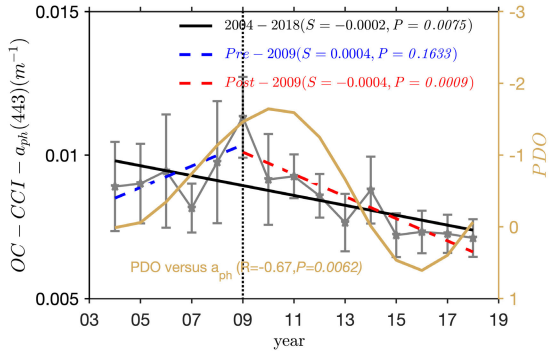


Fig. 7. Satellite time series of phytoplankton absorption coefficients at 443 in the open ocean region of the SCS (18°N–21°N, 114°E–117°E). Gray pentagrams are arithmetic averages, and their associated vertical lines are standard deviations. Black lines, linear regression fit to the entire time series (2004–2018). Blue-dashed lines, linear regressions fit to data only from 2004 to 2009 (pre-2009 includes 2004–2009). Red-dashed lines, linear regressions fit to data only from 2009 to 2018 (post-2009 includes 2009–2018). S, slope of the trend; and P, significance of the trend from Pearson correlation. Yellow lines, PDO index from the National Centers for Environmental Information with Spearman correlation coefficient (R) and significance (P) of the correlation with the $a_{ph}(443)$ data. Low-pass filter (period > 7 years) was applied to the raw PDO time series.

and a_{ph} were slightly lower in the CEs [Fig. 5(d) and (i)]. The most pronounced differences occurred at 100 m depth, where a_p and a_{ph} were substantially lower in CEs, while a_{nap} was slightly higher.

These vertical patterns suggest an eddy-driven modulation of the SCM [80], [81]. In CEs, upward eddy pumping shoals the SCM, concentrating phytoplankton biomass and elevating absorption in shallower layers (25–50 m). Conversely, in ACEs, downward pumping deepens the DCM, shifting peak absorption to deeper layers (75–100 m) [80]. These patterns highlight the role of eddy-induced vertical nutrient and biomass redistribution in shaping the depth-dependent variability of particulate absorption in the SCS.

To quantify the impact of eddies on particulate absorption coefficients, we calculated the relative variability (%) at discrete depths using six wavelengths corresponding to OC-CCI bands (Fig. 6 and Table SII). The relative variability increased with depth and wavelength, resulting from the generally lower absolute values of particulate absorption coefficients at greater depths and longer wavelengths (Fig. 6). For $a_p(443)$, surface waters (5 m) exhibited modest, opposing variability (−5.42% ± 88.17% in CEs versus 3.56% ± 156.99% in ACEs) [Fig. 6(b) and Table SII]. This pattern reversed at 50 m (7.14% ± 61.53% in CEs versus −5.36% ± 57.7% in ACEs) and culminated in a stark contrast at 100 m, with CEs at −13.9% ± 62.24% compared to 11.71% ± 44.81% in ACEs [Table SII and Fig. 6(b)]. A nearly identical vertical progression was observed for $a_{ph}(443)$, while $a_{nap}(443)$ followed a similar pattern but with a consistently greater magnitude of variability in the upper 75 m (Fig. 6). At 100 m depth, $a_{nap}(443)$ decreased within ACEs, while it exhibited a slight increase within CEs. Furthermore, the relative variability was substantially higher at 665 nm than at 443 nm across all depths; for instance, surface variability for $a_p(665)$ reached −19.75% ± 148.22% in CEs versus 12.97% ± 384.68% in ACEs [Table SII and Fig. 6(f)].

Importantly, our results indicate that the influence of mesoscale eddies on particulate absorption was shown considerable vertical heterogeneity. This suggests that eddy-driven modifications to bio-optical properties are complex and nonuniform across the region. This likely reflects the complex and transient nature of eddies, which evolve through distinct developmental stages—growth, maturity, and decay—each with varying physical and biogeochemical signatures [82], [83], [84]. Eddy developmental stages alter nutrient supply and phytoplankton biomass distribution, leading to vertically heterogeneous bio-optical responses. Additionally, eddy intensity, origin, and background environmental conditions (e.g., Kuroshio intrusion) may further modulate their bio-optical impacts [46]. For example, Kuroshio intrusion can induce frontal upwelling, enhancing nutrient supply and phytoplankton growth in the northern SCS [46], thereby modifying vertical bio-optical distributions. Furthermore, the distance of sampling stations from the eddy center may influence the observed patterns, as submesoscale processes near eddy edges can enhance vertical velocities and nutrient supply [85], [86], potentially affecting phytoplankton distribution and bio-optical properties. The current dataset may not fully resolve bio-optical variability across eddy types and life stages, highlighting the need for time-resolved datasets and refined eddy classifications to clarify their influence in the SCS.

D. Long-Term Trends in Particulate Absorption

1) Regime Shift Detection: Over the 2004–2018 period, a significant decreasing trend was identified in the OC-CCI $a_{ph}(443)$ data (slope = −0.0002 m^{-1} yr^{-1}, P = 0.0075, Fig. 7) in the open ocean region of the SCS (18°N–21°N, 114°E −117°E). The in situ data also indicated a decline of similar magnitude, though the trend was only marginally significant [slope = −0.0005 m^{-1} yr^{-1}, P = 0.0966, Fig. 8(a)].

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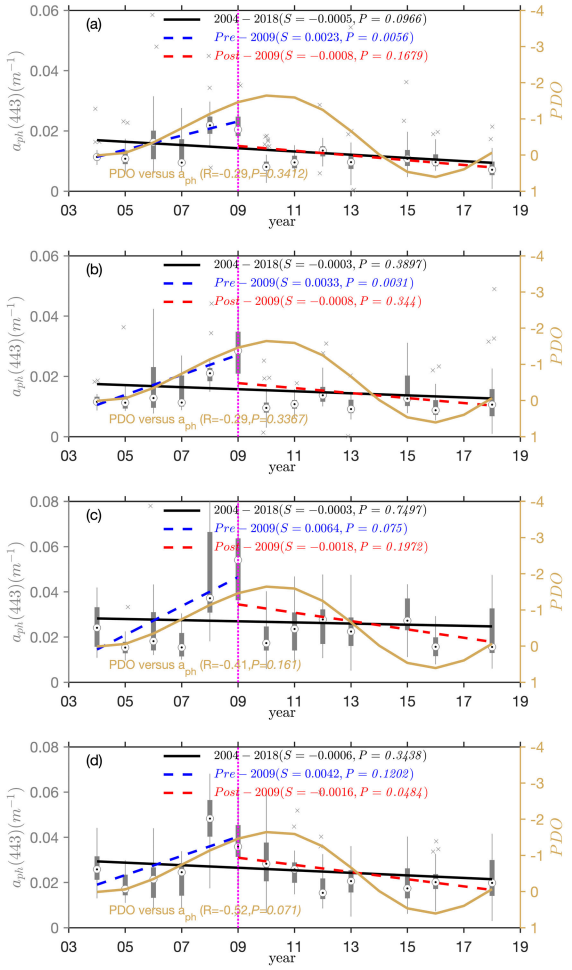


Fig. 8. Time series of phytoplankton absorption coefficients at 443 nm in the depth of (a) 5, (b) 25, (c) 50, and (d) 75 m. Gray dots are arithmetic averages, and their associated vertical lines are standard deviations. Black lines, linear regression fit to the entire time series (2004–2018). Blue-dashed lines, linear regressions fit to data only from 2004 to 2009 (pre-2009 includes 2004–2009). Red-dashed lines, linear regressions fit to data only from 2009 to 2018 (post-2009 includes 2009–2018). S, slope of the trend; and P, significance of the trend from Pearson correlation. Yellow lines, PDO index from the National Centers for Environmental Information with Spearman correlation coefficient (R) and significance (P) of the correlation with the $a_{ph}(443)$ data. Low-pass filter (period > 7 years) was applied to the raw PDO time series.

Furthermore, no significant trends were observed in the subsurface in situ data [Fig. 8(b)–(d)], likely due to limited data availability. Following Cheng et al. [87] (2023), we partitioned our time series into pre- and post-2009 periods. Trend analysis of in situ observed $a_{ph}(443)$ revealed a pronounced statistically significant shift after 2009. Specifically, a significant increasing trend was observed from 2004 to 2009 [slope = 0.0023 m^{-1} yr^{-1}, P = 0.0056, Fig. 8(a)]. In contrast, this trend reversed after 2009, with the data from 2009 to 2018 showing a nonsignificant decline [slope = −0.0008 m^{-1} yr^{-1}, P = 0.1679, Fig. 8(a)].

In the subsurface layers, the significant pre-2009 increasing trend was also evident at 25 and 50 m [Fig. 8(b) and (c)]; however, the post-2009 decreasing trends at these depths were not significant and only suggestive of a decline [Fig. 8(b) and (c)]. Conversely, at 75 m, the post-2009 decreasing trend was significant the decreasing trend after 2009 was significant (slope = −0.0016 m^{-1} yr^{-1}, P = 0.0484), while the pre-2009 increasing trend was not statistically significant (slope = 0.0042 m^{-1} yr^{-1}, P = 0.1202) [Fig. 8(d)].

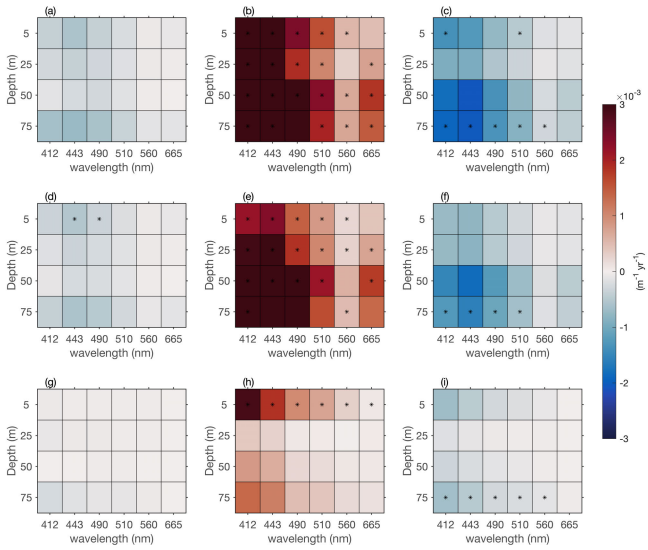


Fig. 9. (a)–(c) Linear trends of total, (d)–(f) phytoplankton, and (g)–(i) nonalgal particulate absorption coefficients. Shading color represents the slope of the trend. Stippling in (a)–(i) indicates where the linear trends are significant at the 90% confidence level according to the Student's t-test. (Left) Linear regression fit to the entire time series (2004–2018). (Middle) Linear regressions fit to data only from 2004 to 2009 (pre-2009 includes 2004–2009). (Right) Linear regressions fit to data only from 2009 to 2018 (post-2009 includes 2009–2018).

2) Vertical and Spectral Patterns: Our analysis was extended to include the absorption coefficients of total particulates, phytoplankton, and nonalgal particles across multiple OC-CCI wavelengths (412, 443, 490, 510, 560, 665 nm) and discrete depths (5, 25, 50, 75 m) (Fig. 9). For either a_p or a_{ph} or a_{nap} shown a no significant decreasing trend over the entire time series, except for a_{ph} at 443 and 490 nm in 5 m. Over the entire time series, no significant decreasing trends were identified for a_p or a_{nap} at any wavelength or depth [Fig. 9(a) and (g)]. A significant decreasing trend was only found for a_{ph} at 443 and 490 nm in the surface layer (5 m) [Fig. 9(d)]. The significant pre-2009 increasing trend was evident for a_p or a_{ph} at nearly all wavelengths and depths [Fig. 9(b) and (e)]. In contrast, for a_{nap}, this trend was only observed at the surface (5 m). After 2009, a significant decreasing trend was identified primarily at 75 m depth for a_p, a_{ph}, and a_{nap} [Fig. 9(c), (f), and (i)]. Furthermore, for a_{ph}, the rate of the pre-2009 increase grew larger with depth, and similarly, the rate of the post-2009 decrease became more pronounced at greater depths [Fig. 9(e) and (f)].

E. Climate Effect on Particulate Absorption

1) PDO Influence: Regional phytoplankton absorption coefficients exhibit both short-term oscillations and long-term trends. We tested the hypothesis that the short-term (interannual to decadal) variability in $a_{ph}(443)$ is influenced by leading climate oscillations, particularly the PDO. Indeed, OC-CCI $a_{ph}(443)$ shown a strong negative correlation with the PDO index in the SCS (R= −0.67; P = 0.0062; Fig. 7), suggesting that its decadal variability is modulated by the PDO.

Mechanistically, the enhanced trade winds during 2004–2009 generated downwelling Rossby waves and

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upwelling Kelvin waves, resulting in a deeper thermocline and elevated SST in the western Pacific [87]. The resultant increase in ocean stratification and a deeper nutricline favor a shift in phytoplankton community structure toward smaller species [88], [89]. Because smaller phytoplankton have a higher specific absorption coefficient due to their greater surface-to-volume ratio [12], this community shift may directly contribute to the observed increase in $a_{ph}(443)$ (Fig. 8). After 2009, the opposite conditions prevailed: strong westerly wind anomalies induced downwelling Kelvin waves and upwelling Rossby waves propagating eastward and westward, respectively, promoting larger phytoplankton species and corresponding decreases in $a_{ph}(443)$.

2) ENSO Comparison: The ENSO is another major driver of interannual oceanic variability, influencing global phytoplankton dynamics through changes in SST, wind patterns, and thermocline structure [90], [91], [92]. In the SCS, a significant negative correlation has been reported between the MEI and chlorophyll-a concentrations [90], [93]. The strongest ENSO-related chlorophyll-a concentration anomalies typically occur in regions such as the southeastern Vietnam upwelling zone and along the Chinese coast, where El Niño events weaken monsoon-driven mixing and nutrient supply. In the present study, however, the relationship between particulate absorption coefficients and the MEI was weak (Fig. S4), suggesting that particulate bio-optical properties may respond only indirectly to ENSO forcing. Conversely, a moderate negative correlation was observed between particulate absorption coefficients at 75 m and the PDO, suggesting that PDO may influence the variability of $a_{ph}(443)$.

IV. Summary

Understanding long-term bio-optical variability in marginal seas like the SCS is critical for linking regional ocean dynamics to global biogeochemical cycles. The SCS, influenced by strong monsoon forcing, Kuroshio intrusion, and energetic mesoscale eddies, provides a unique natural laboratory to examine how physical processes shape optical and biological properties in oligotrophic waters. This study explores hyperspectral particulate absorption characteristics over 13 years, revealing how shifts in upper-ocean circulation and large-scale climate modes modulate light absorption by phytoplankton and detrital particles. Our results show that mesoscale eddies exert vertically heterogeneous effects on particulate absorption, reflecting complex interactions between eddy dynamics, nutrient availability, and phytoplankton distribution. The findings demonstrate that even subtle variations in particulate absorption can reflect long-term ecological transitions associated with decadal-scale climate oscillations, highlighting the sensitivity of bio-optical properties to climate-driven environmental change. Moreover, the research underscores the importance of sustained field observations for validating and improving ocean-color remote sensing algorithms, particularly in optically complex waters. By combining long-term in situ observations with an analysis of mesoscale and climate-scale variability, this work advances our understanding of bio-optical responses to environmental forcing in the SCS and establishes a foundation to refine ocean-color products in the SCS. It is also important to note that the in situ data were collected exclusively during summer; therefore, the relationships identified (e.g., SST-absorption correlations) may not fully represent annual variability. Future multiseasonal campaigns are needed to validate and extend these findings across the full seasonal cycle.

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Physical–biochemical data collected onboard used in this study are available upon request. Figures were made with MATLAB version 9.12.0.1884302 (R2022a), available under the MATLAB license at https://www.mathworks.com/products/MATLAB.html (MATLAB, 2022).

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